flysdown: a liveADS-BandAIS dashboard with geofence projection
How the system works, why it is built this way, and where every fact in it came from

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Keywords: ADS-B, AIS, geofencing, dead reckoning, closest point of approach, serverless edge computing, live cartography.

Version 1.1, 17 September 2026. Live at flysdown.pages.dev (target domain flysdown.jaronwilson.dev). Source: Jaron-Wilson/flysdown.

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Abstract

flysdown plots live aircraft and live ships on one map and runs a detection engine over them that answers one question: if this target holds its current track and speed, does it end up somewhere it should not be, and how long have we got. Aircraft positions come from community ADS-B aggregators, ship positions from a national AIS service, and restricted airspace from the FAA's own published dataset. It runs as static files plus two edge functions on Cloudflare's free tier, plus one background poller on an ordinary connection.

This paper documents the data sources and their quirks, the architecture, the detection mathematics, the airspace pipeline, the visual encoding and the verification. One problem shaped the design more than any other: the community ADS-B services rate-limit by IP address, and server-less edge platforms egress from addresses shared with every other customer, so the obvious architecture does not work. Section 4 measures that and describes the fix.

Claims are tagged by provenance. [measured] means obtained by instrumenting this system on 16 to 17 September 2026. [documented] means from a provider's own documentation, cited in Section 13. [standard] means from a published standard or regulation. Section 10 collects the load-bearing claims in one table.

1. Scope

The system shows live aircraft (ADS-B) and ships (AIS) on one dark map, aircraft colored by altitude and ships by whether they are under way, and runs a rules engine on every update: already inside a restricted zone, projected to enter one with a time to the boundary, emergency transponder codes, rapid descent, and orbiting or holding. It carries the FAA's published prohibited areas with their real boundaries and limits, plus the statutory Washington DC Special Flight Rules Area and the two standing Disney restrictions. An operator can draw further watch zones (circle or polygon), set their floor, ceiling and whether they apply to aircraft, ships or both, and export or import them as GeoJSON. By default it tracks only the area on screen, or an operator can pin circles and boxes that keep loading while the map is scrolled anywhere else. Selecting an aircraft draws the track it has been observed flying plus its published origin and destination as great circles. Each feed pauses indepen-dently, every contact is timestamped, and the interface reports which path served its data and how old that data is.

It is not a navigation tool and says so on every screen. The projection is straight-line dead reckoning: no turns, no wind, no flight plan, no con-troller instruction. NOTAM activation is not modeled, so a restricted area that is cold today is still drawn.

The system is 6,627 lines across browser modules, edge functions, the shared fetch layer, the relay, tooling and tests [measured], with no build step and no framework. MapLibre GL JS is vendored as one 954 KB file so the page does not depend on a third-party script host at runtime.

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2. The source data

2.1 ADS-B

ADS-B is cooperative surveillance: an aircraft derives its own position, usually from GNSS, and broadcasts it unprompted with identity, alti-tude, velocity and status. The 1090 MHz Extended Squitter link and its message formats are specified in RTCA DO-260B and ICAO Annex 10 Volume IV [standard]. Anyone with an antenna and a software defined radio can decode it, which is why a volunteer receiver network exists and aggregate live data is free.

Figure 1. The dashboard over Washington. The left rail is tabbed (Overview, Filters, Areas), the map carries the FAA prohibited areas and the statutory DC Special Flight Rules Area, and the right rail holds alerts and the selected target. 104 aircraft were in the covered area when this was captured.

What arrives here is the JSON that the readsb decoder and its tar1090 interface produce, which aggregators serve directly; field seman-tics are in the readsb JSON reference [documented]. Three quirks are load-bearing:

1. alt_baro is not always a number. For a target on the ground it is the string "ground". Unchecked arithmetic produces NaN altitudes that poison every downstream comparison. The normalizer converts it to an explicit onGround: true at altitude zero.
2. track is often absent: 27 of 115 aircraft over Washington in one sample [measured], typically surface or otherwise limited targets. The normalizer falls back to true_heading, then mag_heading, but never to nav_heading, which is the heading the autopilot is steering towards rather than the one being flown.
3. dbFlags is a bitmask, not a value to compare: bit 1 military, 2 interesting, 4 privacy ICAO address, 8 limited data display. These become four booleans.

Emergency transponder codes carry a separate channel of meaning in squawk: 7500 unlawful interference, 7600 radio failure, 7700 general emergency, per the FAA Aeronautical Information Manual [standard]. The engine treats 7500 and 7700 as critical and 7600 as serious, and also reads the ADS-B emergency field.

2.2 Aggregators, and their terms
Three keyless aggregators were evaluated, and their terms changed the code, so they are quoted rather than paraphrased.

adsb.lol publishes API data under the Open Data Commons Open Database License 1.0 and states "The API is available to everyone" [docu-mented]. ODbL is an attribution license, so the map credits it.

adsb.fi is stricter. It states that "The public endpoints are rate limited to 1 request per second", that invalid requests (400, 401, 403, 404, 429) count towards restrictions and "may trigger temporary IP blocks", that the data "is for personal, non-commercial use only", and that users "must cite adsb.fi and include a link to our home page" [documented]. Three consequences: the relay spaces upstream requests 1.1 seconds apart rather than bursting one per region; adsb.fi is never called from the edge, where it returns a 403 bot challenge every time (Section 4) and those 403s would accumulate strikes against a shared address for a call that has never once succeeded; and the map attribution cites adsb.fi with a link to its home page. adsb.fi also documents v2/lat/lon/dist as deprecated in favor of v3, which returns the same ac -keyed shape as its other v2 endpoints [documented]; the system now uses v3, with the response shape re-verified before the switch [measured].

OpenSky Network serves state vectors as positional arrays in SI units [documented]. It works from an ordinary connection (0.56 s, with x-rate-limit-remaining: 399 confirming the documented anonymous allowance) [measured] but is unreachable from the edge, so it is re-lay-only.

2.3 AIS

AIS is the maritime analog: transponders broadcast position, course, speed and identity on VHF, with encodings specified in ITU-R Recommendation M.1371 [standard]. Three details matter:

Sentinel values. Course over ground 360, heading 511 and speed over ground 102.3 mean "not available", not a measurement. Unhandled,
a fleet appears to steam due north at 102 knots. In one 220-vessel sample, 10 had no usable course and 40 no usable heading [measured].

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Ship type is a coded integer with structural ranges: 30 fishing, 35 military operations, 36 sailing, 37 pleasure craft, 50 to 59 special craft (pilot, tug, search and rescue), 60 to 69 passenger, 70 to 79 cargo, 80 to 89 tanker [standard].
ETA is packed into 20 bits: month in 4, day in 5, hour in 5, minute in 6 [standard]. The decoder unpacks it and rejects impossible combi-nations. Validated live: raw 625728 decodes to 17 September 17:00 UTC, a plausible arrival the day it was read [measured].

Live AIS is far harder to get without a commercial contract than ADS-B. The one genuinely keyless live source found is Fintraffic's Digitraffic marine service, publishing locations and vessel metadata as separate endpoints under CC BY 4.0 [documented]. Coverage is the Baltic and Gulf of Finland: the full feed carried 1,157 vessels spanning latitude 56.9 to 65.8 and longitude 17.0 to 35.3 [measured]. That limit is stated in the interface, with a region switcher and an explicit message when the view is outside coverage.

Digitraffic has one requirement that costs an afternoon if unread: compression is mandatory, and "If compression is not allowed in the request, the service returns error code 406 " [documented]. This was diagnosed the hard way. The endpoint returned 406 for every Accept value tried, including application/json, application/geo+json and */*, and 200 as soon as Accept-Encoding: gzip was present [measured]: the header that looked like the problem was not the problem. Digitraffic also asks callers to send a Digitraffic-User header, noting that "Using the header increases the amount of requests you can make" [documented], so the proxy sends it.

Position and identity arrive separately (/locations keyed by MMSI, /vessels for names, call signs, types, dimensions, destinations). The proxy merges them by MMSI and caches metadata for 30 minutes against 12 seconds for positions, because a ship's name changes less often than its position.

2.4 Flight routes
ADS-B broadcasts identity, position, altitude and velocity. It does not broadcast where a flight came from or where it is going, because the air-craft is not telling you its schedule, only its state. Origin and destination therefore come from a fourth source: adsbdb, which resolves an airline callsign to origin and destination airports with coordinates, built on volunteer flight route data and published under the MIT licence with its data contributors credited [documented].

Three things about using it. Most general aviation and military callsigns have no scheduled route at all, and the API answers 404 for them, which is a normal result rather than an error: the interface says "no scheduled route for this callsign" instead of looking broken. A callsign's route does not change during a day, so lookups are cached at the edge for six hours, and misses for one hour, which means one upstream re-quest per callsign per six hours no matter how many people click on it. And unlike the ADS-B aggregators in Section 4, adsbdb answers the edge without complaint, verified from the deployed Worker [measured], so this one needs no relay.

3. Architecture

browser (public/) Cloudflare edge (functions/) upstreams
app.js orchestration /api/aircraft adsb.lol
js/feeds polling, history ---> normalize, quantize, cache ---> (adsb.fi and
js/detect rules engine relay lookup, stale fallback OpenSky are
js/geo geodesy relay-only)
js/zones zone store ---> /api/vessels ---> Digitraffic
js/map MapLibre layers merge positions + metadata
js/draw zone drawing
js/ui panels D1 <--- tools/relay.mjs (ordinary IP)

The browser is plain ES modules, no framework, no build step. That was decided explicitly: a live map is overwhelmingly client-side render-ing work, and the only thing a server must do is proxy two feeds, normalize them and hide rate limits behind a shared cache. A JVM or Python service would need an always-on host for no functional gain. The detection engine and its geodesy are pure functions with no DOM or map de-pendency, specifically so they can move into a scheduled Worker or another language later without rewriting the rules.

The edge layer is two Cloudflare Pages Functions [documented], one per feed, plus a relay endpoint. They exist because the upstreams send no CORS headers so a browser cannot call them; secrets stay server-side; parameters can be quantized so many viewers collapse onto one cache entry; and one cache entry serves every viewer instead of each viewer generating upstream load.

Normalization is a contract. Both endpoints emit one schema whatever answered, so an aircraft is the same object whether it came from adsb.lol's ac array, adsb.fi's v3 ac array or OpenSky's positional arrays in SI units (conversions in one place: 3.28084 ft/m, 1.943844 kt per m/s, 196.8504 ft/min per m/s). The same code runs at the edge and in the relay, both importing shared/adsb.js, so a relayed snapshot cannot drift in shape from a directly fetched one.

Quantization and caching. Latitude and longitude are rounded to 0.1 degrees (about 6 nautical miles) and radius to 25 nautical mile steps, so two people looking at the same city produce the same region key and share one upstream request. Responses are cached in the Cloudflare Cache API [documented] for 6 seconds (aircraft) and 12 (vessels) at the edge with max-age=0 for the browser, so the browser always asks and the edge answers most asks without touching an upstream. A second, longer entry holds the last known good answer for 5 minutes, which is what makes the degradation in 4.4 possible.

4. The problem that shaped everything: egress addresses

4.1 Symptom and measurement
The first deployment worked perfectly locally and returned nothing in production: 179 aircraft over Washington in local development, HTTP 502 with all three upstreams refusing when deployed [measured]. Rather than guess, a temporary diagnostic endpoint was deployed that called each upstream from the edge and reported status, timing, headers and a body snippet. From the Cloudflare IAD colocation [measured]:

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UPSTREAM FROM THE EDGE FROM AN ORDINARY IP

adsb.lol 200 in 463 ms when it works, HTTP 429 (nginx) on most attempts works every time

adsb.fi HTTP 403 in 4 ms, body is a Cloudflare bot challenge page works every time, richest fields

OpenSky HTTP 522 after 19,533 ms, then 4 timeouts in 4 attempts 200 in 0.56 s

Fresh-fetch success rates from the edge: 1 in 8 with no retry logic, 3 in 8 once a jittered retry on 429 was added, 1 in 6 when re-measured an hour later [measured]. The variance is itself informative: the limiter is reacting to the aggregate traffic of everything sharing that address, not to this site.

4.2 Diagnosis
The aggregators rate-limit by IP, the only practical identifier for anonymous users. A serverless function has no address of its own; it egresses from addresses shared with every other customer on the platform in that location, so requests compete against the world for a per-address budget. adsb.lol's documentation notes that its limits are dynamic according to load [documented], which fits the observed variance.

Two dead ends were eliminated before redesigning. Browser-direct fetching: none of the aggregators return Access-Control-Allow-Origin, so a page cannot call them from the user's own address [measured]. A different source: six further candidates were probed and all failed: api.adsb.one (403 even from an ordinary IP), api.adsb.im and api.theairtraffic.com (unresolvable), data.adsb.fi (401), globe.adsb.fi (403) and api.planespotters.net (404) [measured]. The constraint is the address, not the source, so swap-ping sources cannot fix it, and an OpenSky account would not help either because OpenSky does not answer the edge at all.

4.3 Graceful degradation as the floor
Whenever the relay is not running, the endpoint serves the last known good answer from the 5 minute cache entry, annotated stale: true with its age, and the interface says so in three places: the feed chip, the status bar and a map banner. Verified in production by polling one re-gion repeatedly: four refusals, then 323 aircraft fresh, then the same 323 marked stale at 17 and 29 seconds old [measured]. The property that matters is honesty: a dashboard silently showing five minute old positions as current is worse than one showing nothing, because the viewer cannot tell.

4.4 The relay
The fix is to move the fetch to an ordinary address. tools/relay.mjs runs anywhere with a normal connection, and each cycle it asks the site which regions people are looking at (GET /api/relay, bearer token), fetches those from the aggregators preferring adsb.fi for its richer fields while spacing requests 1.1 seconds apart, and pushes each result back (POST /api/relay) into Cloudflare D1. /api/aircraft reads a fresh snapshot from D1 before trying any upstream, so when the relay runs the edge never contacts an aggregator at all: more reliable, and considerably politer than retrying into a rate limiter.

Demand is recorded by the aircraft endpoint itself: a cache miss writes the canonical region key to a wanted table. The relay therefore fol-lows the map rather than polling a fixed list of cities, a newly viewed area is covered within one cycle, regions unviewed for ten minutes are swept, and the list is capped at five.

Covering snapshots. The region key includes the radius, which changes with zoom, so panning produced regions the relay had not yet cov-ered and the first request for them returned nothing. Requests are now answered by any snapshot that fully contains the requested area: the endpoint picks the tightest containing snapshot and filters its aircraft to the requested radius. Verified in production: a 25 nautical mile request served from the 75 nautical mile snapshot, filtered from 181 aircraft to 71, and an off-center 50 nautical mile request from the same snapshot giving 117 [measured].

Write budget. D1's free allowance is 5 million rows read and 100,000 written per day [documented]. One row per region per cycle, five re-gions on an 8 second interval, is about 54,000 writes per day. Two throttles keep clear of the ceiling: demand recording reads the existing timestamp and only writes if it is over two minutes old (a region stays listed for ten minutes, so refreshing sooner is pure write volume), and housekeeping deletes run on about one poll in twenty. D1 also caps a Worker invocation at 50 queries on the free plan [documented]; this end-point uses at most three.

When the relay stops, nothing breaks: the endpoint falls back to the aggregators, then an older relay snapshot, then the last known good cache entry, then a 502 with a summarized reason. The banner distinguishes "the aggregators are rate-limiting the edge" from "the relay has stopped pushing", because those have different remedies.

5. The detectionengine

5.1 Geodesy
Everything geometric happens on a sphere of radius 3,440.065 nautical miles using standard haversine distance and great-circle destination formulas [documented], with distance in nautical miles, speed in knots, altitude in feet, vertical rate in feet per minute and bearings in degrees true. The unit choice is not aesthetic: it makes time, distance and speed relate without conversion factors, removing a category of error from the projection. A spherical model is accurate enough here; the worst-case error against an ellipsoid over 250 nautical miles is a fraction of a per-cent, far below the uncertainty in extrapolating from one instantaneous velocity sample.

One consequence caught a bug, in the test rather than the code. A test placed an aircraft 60 nautical miles due west of a zone and tracked it 090, expecting a hit; it missed. Great circles converge, so the bearing back along the great circle differs from 090 by roughly the longitude difference times the sine of the latitude, about 0.8 degrees at latitude 38.9, which over 60 nautical miles is 0.85 nautical miles of cross-track error: against a 0.25 nautical mile zone, a clean miss. The fix was to compute the true initial bearing rather than assume the reciprocal.

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5.2 Projection
Take the current position, track and ground speed, step forward in time, and extrapolate altitude linearly from the vertical rate. It models nothing else, because the question a geofence warning needs answered is "if nothing changes, what happens", not "what will this aircraft actually do".

Altitude extrapolation is clamped at ground level. A constant 4,000 feet per minute descent extrapolated ten minutes forward is 40,000 feet lower, which for most aircraft is underground, and an unclamped negative altitude silently falls below every zone floor, so a steeply descend-ing aircraft would stop alerting exactly when it became most interesting. Alongside the clamp is an explicit time-to-ground: if a target reaches the surface at its current rate before the horizon, the projection stops there rather than flying a landed aircraft across the map. Both are tested.

5.3 Finding the first intersection efficiently
For each candidate pair the engine finds the first moment the target is inside the zone both horizontally and vertically within the horizon (default 10 minutes, adjustable 2 to 20). Naively that is hundreds of targets times a dozen zones times hundreds of steps, five times a minute. Three things make it cheap.

A reachability prefilter. Maximum travel is speed times horizon; if the distance to the zone center less its radius exceeds that, the zone is skipped with no stepping at all, which eliminates almost every pair.

A step tied to the zone, not the clock. The step is max(0.05, min(2, zoneRadius / 2, 0.5)) nautical miles, converted to seconds from the target's speed and floored at one second. A fixed 15 second step sounds reasonable and is a real bug: at 500 knots that is 2 nautical miles per step, which steps clean over P-56B (one nautical mile radius) without ever sampling inside it. There is a test using a 0.25 nautical mile zone and a 550 knot target.

Bisection to refine. Stepping establishes a bracket; 12 iterations between the last clear and first inside sample narrow the crossing to roughly one four-thousandth of a step, which is what lets the interface say "reaches P-40 Thurmont in 7m 52s" rather than rounding to the sample interval.

Containment is analytic for circles (distance to center) and ray casting for polygons (count edge crossings; odd is inside), verified against two known truths: the White House is inside P-56A, Dulles is not.

5.4 Rules and severity

RULE CONDITION SEVERITY

Inside a zone Horizontally inside, between floor and ceiling Prohibited/TFR critical; restricted/custom serious

Projected entry First intersection within the horizon Base rank, reduced one step beyond 2 min, two be-
yond 6

Emergency squawk 7500 or 7700 Critical

Emergency squawk 7600 Serious

Emergency flag ADS-B emergency field set Critical

Rapid descent Below -4,000 ft/min, or -3,000 ft/min under 10,000 ft Warning; serious when low

Orbit or hold Over 270 degrees cumulative turn inside a 12 NM footprint over
150 s

Notice

Close approach
(vessels)

Projected CPA within the limit (default 1 NM) and TCPA inside 30
min

By CPA band, reduced one step beyond 15 min

Severity is computed, not stored: zone kind gives a base rank and the projected time reduces it, so an aircraft escalates from warning to seri-ous to critical as it closes without any state being kept. Alert identifiers are stable (targetId:rule:zoneId) so the interface tracks an alert across updates instead of flashing a new one every five seconds.

Orbit detection needs history, so the browser keeps up to 150 positions per target over a 10 minute window; the rule sums absolute heading changes and requires the bounding circle of those positions to stay small, separating a hold or survey orbit from an aircraft simply turning en route. Rapid descent uses two thresholds because 3,500 feet per minute at cruise is routine and the same rate at 2,000 feet is not.

5.5Advisory zones, or not burying the signal
The first version produced 55 simultaneous alerts over Washington, about 50 of them variations of "an airliner is inside the Washington DC Special Flight Rules Area" [measured]. True and useless: the SFRA is a 30 nautical mile ring inside which transiting requires a flight plan, a discrete code and two-way radio, so essentially all traffic in it is authorized. Zones therefore carry an advisory flag, default true for special-flight-rules zones: such a zone is still drawn, still tested and still reported in a selected target's zone checks, but never raises an alert. The same view then produced 3 alerts, all genuine: a projected entry into P-40 at Camp David and two into P-56A over the National Mall [mea-sured]. A detector that fires on authorized behavior trains its operator to ignore it.

5.6 Vessel close approach
Every rule above looks at one target at a time, which cannot see the risk that matters most at sea: two ships converging on each other. That needs a pairwise pass, and the marine standard for it is the Closest Point of Approach.

Reduce the pair to relative motion. Convert both positions into a local tangent plane in nautical miles, take the relative position vector r and the relative velocity v (each vessel's speed over ground resolved onto its course over ground), and the time of closest approach is where the range stops shrinking:

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TCPA = -(r. v) / |v|^2 CPA = | r + v * TCPA |

A negative or zero TCPA means the pair has already passed its closest point or is opening, in which case the closest approach is simply the present range. A relative speed near zero means two ships holding station on each other, so the range is not going to change either. Both are re-turned as "not closing" rather than as a spurious prediction.

This is the quantity an ARPA radar computes, and the alarm model is the same too: IMO's performance standards for automatic radar plotting aids require alarms on "a preset minimum acceptable passing distance (CPA) and a preset advance warning time (TCPA)" [standard]. Both are exposed here, the CPA limit as a slider defaulting to 1 nautical mile and the horizon at 30 minutes, because the right limit depends entirely on the water: a mile is generous in a traffic separation scheme and tight in the open sea.

Two guards keep the output worth reading. Both vessels must be making at least 1 knot and neither may be reporting anchored or moored, be-cause a harbor is full of ships lying within a cable of each other and none of it is a risk, the same lesson as the advisory zones in 5.5. And pairs are screened through a coarse spatial grid sized to the 20 nautical mile screening range, so the pass is linear in practice rather than quadratic in the contact count. Each pair yields exactly one alert, keyed on both identifiers in a stable order, so the same encounter is never reported twice from opposite ends.

Measured against live Baltic traffic, 275 vessels produced 4 alerts at the 1 nautical mile limit and 27 at 5, the tightest being two ships projected to pass 0.32 NM apart in 2 minutes 14 seconds [measured]. Each pair is drawn on the map as a line between the two vessels, colored by severity.

Figure 3. Vessel traffic in the Gulf of Finland with two close-approach warnings in the alert rail. Under way vessels are orange, moored and anchored ones gray; the latter are excluded from the pairwise pass, since a harbor is full of ships lying a cable apart at zero knots. The alert cards carry each pair's projected passing distance and the time to it.

5.7 A worked example
From the live system: N9287Y, west of Camp David at 5,000 feet, with track and ground speed from ADS-B. P-40 Thurmont is prohibited, surface to 5,000 feet [documented], geometry from the FAA. The prefilter keeps the pair because the aircraft can reach it inside the horizon; stepping at 0.5 nautical mile granularity finds the first sample inside; bisection refines the crossing to 56 seconds; the projected altitude at that moment is inside the band. Base severity for a prohibited area is critical and 56 seconds is inside the 2 minute imminent band, so no reduction applies. The alert reads "Projected entry: P-40 Thurmont, on current track N9287Y reaches P-40 Thurmont in 56s, 5000 ft", the zone outline thickens, and a dashed projection line is drawn to the predicted entry point [measured].

6. Airspace data

6.1 Using the FAA's own geometry
The first zone file was hand-drawn: P-56A as a rectangle over the National Mall, P-40 as a one mile circle, the DC flight restricted zone as a 15 nautical mile ring. It was labeled approximate and it generated false positives, because a rectangle over the Mall includes the Potomac corridor that Reagan National arrivals actually fly, which the real prohibited area is shaped to exclude.

The FAA publishes Special Use Airspace as a queryable ArcGIS feature service through its Aeronautical Information Services open data portal [documented]. Querying for prohibited areas returns 13 features with real boundaries and published limits [measured]. The same service holds 555 restricted areas, 718 military operations areas, 212 warning areas, 39 alert areas and 5 of type D [measured]; only prohibited areas are ingested, because they are permanent and unconditional, whereas most restricted areas and MOAs are active only by NOTAM or schedule and drawing them as always hot would mislead.

tools/fetch-zones.mjs performs the ingest and is committed, so the data is reproducible; the FAA reissues the dataset on the 56 day chart cycle, so the script is re-run rather than its output edited. Three mapping details: LOWER_VAL and UPPER_VAL are strings with separate code fields where SFC means surface, normalized to numeric feet; the country field reads "UNITED STATES", not "USA", and an initial filter

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comparing against USA silently dropped all 13 features, which is why the script now prints every zone it wrote with its vertex count; and P-56 appears twice under one name, disambiguated by geometry, the small circle at 38.9214, -77.0669 being section B over the Naval Observatory and the Mall polygon section A.

6.2 Simplification
The FAA ships circular areas as densely sampled polygons: P-56B, a one nautical mile circle, arrives as 6,285 coordinate pairs, as does P-40 [measured], and 13 zones came to roughly 520 KB of JSON. The ingest applies Douglas-Peucker simplification [standard] at a tolerance of 0.0004 degrees (about 44 meters) and rounds to five decimal places. P-56B becomes 17 points, P-40 33, P-56A 12, and the whole file includ-ing hand-added zones is 16 zones, 506 polygon vertices and 18.6 KB [measured]. The difference is not visible at any zoom the map offers, and the simplified geometry is still tested for containment.

6.3 Zones defined by regulation, and one that was dropped
Two zones are built from their legal definition and are therefore exact rather than approximate. The DC Special Flight Rules Area is defined in 14 CFR part 93 subpart V as the airspace within 30 nautical miles of the Reagan National VOR/DME [standard], so a circle is the true shape. The prohibited areas are established under 14 CFR part 73 [standard], the authority behind the FAA geometry. The two Disney restrictions are permanent TFRs with published centers, 3 nautical mile radius, surface to 3,000 feet AGL.

The DC flight restricted zone was dropped rather than approximated: its real boundary is an irregular coordinate-defined polygon, the dataset does not contain it, and a circular approximation was both wrong and the largest single source of false alerts. Shipping nothing was more hon-est than shipping a bad shape.

Altitude datum caveat. Floors and ceilings are compared against barometric altitude in feet MSL, but several real restrictions are published in feet AGL (P-40 to 5,000 AGL, Disney to 3,000 AGL). Those zones are flagged agl: true, and the comparison error is the terrain elevation, which the system does not model. This is stated in the zone metadata and in the interface.

7. Visual encoding

7.1 Color assigned by role, then validated
Color here encodes data, so it was assigned by the job each color does and then checked with a contrast and color-vision validator against the page's actual dark surface rather than judged by eye.

The chrome is jaronwilson.dev and jaronwilson.org's own palette and type, verbatim, so the three sites read as one person. That moved the data's surface from #141416 to #201d16, so every palette below was re-validated against it. One rule falls out: the brand accent
(#d97a4a) is seven units of color difference from the vessel orange, inside the range where a viewer cannot tell them apart, so the accent stays in the chrome and never appears on the map, in the legend or in the data panels, and no data color appears in the chrome.

Altitude is a magnitude, so it gets an ordinal ramp on a single hue, monotone in lightness, dark low and light high: #184f95, #256abf, #3987e5, #6da7ec, #9ec5f4, #cde2fb for bands below 2,500 ft, to 10,000, 20,000, 30,000, 40,000 and above, passing all four ordinal checks with the darkest step at 2.08:1 against the surface [measured]. A rainbow ramp, which several trackers use, was rejected: hue carries no order, so two colors cannot be ranked without a legend.

Aircraft against vessels is identity, so categorical hues: blue #3987e5 and orange #d95926, a pair that passes all-pairs color-vision sepa-ration [measured]. Gray #898781 is reserved for "no data, not moving or stale", never an identity color, and always labeled.

Alerts are status, using a reserved four-step palette (#d03b3b critical, #ec835a serious, #fab219 warning, gray notice) never reused for a data series. Red against green is inherently weak under deuteranopia, measured at a difference of 4.1 [measured], which is why every alert pairs its color with a glyph and the severity word: status color never carries meaning alone here.

Zone kind is identity with a safety-critical failure mode, so it is encoded twice. The four hues clear the normal-vision floor comfortably (worst pair 24.6) but sit in the color-vision warning band (7.2) [measured], which is only acceptable with a second channel, so zone kind is also car-ried by outline dash: prohibited solid, TFR short dash, restricted long dash, special flight rules dash-dot, custom dotted. Because MapLibre's line-dasharray cannot be data driven [documented], this is one line layer per kind with a filter: more code, but the redundant channel sur-vives. Every zone is also directly labeled.

Icons are generated at runtime on a canvas, one pre-colored image per altitude band and vessel state, with a dark casing stroke so a light icon stays legible over a light coastline; the alternative, one white icon tinted by data, needs signed distance fields and gives soft edges at small sizes.

7.2 Only what is on screen, and how old it is
The upstreams take a center and radius, so the smallest circle covering a rectangular viewport always includes targets outside it. Those are fetched (they cost nothing) but filtered out of the map, counts, alerts, trails and table, so "105 aircraft in view" is literally true. The coverage ring is drawn only when the 250 nautical mile upstream cap actually cuts into the view. Targets are dropped when they leave the covered area, both on a successful poll and immediately on a view change; without the second rule, switching from Washington to the Gulf of Finland displayed 241 aircraft where the feed reported 1 [measured].

That default can be overridden by pinning up to four tracking areas, drawn as circles or boxes. Pinned areas replace the viewport as both the query set and the display filter, so the feeds keep loading them while the map is scrolled somewhere else entirely, and their contacts stay in the counts and the alerts even when off screen. Because the upstreams take a center and a radius, a box is queried by its bounding circle and then filtered back to the rectangle for display. Areas persist in localStorage, and the map says so when none of them is on screen. Each feed also pauses on its own, so the aircraft picture can be frozen for inspection while the ships keep moving.

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Every contact carries an age that sums two things: how long ago the receiver network last heard from it (seen_pos, or the AIS report time-stamp) and how long ago this system fetched that answer. Reporting only the first would let a five minute old relay snapshot claim every con-tact in it was two seconds old. The age appears in the detail panel, in the tooltip past 20 seconds, as a table column, and on the map as opac-ity: contacts fade from full to 30 percent between 45 and 240 seconds, so a stale picture looks stale.

7.3 Finding your way around
The first layout put every control on screen at once, in one long scrolling rail. It was complete and it was overwhelming, especially for a visitor arriving from a link with no idea what they were looking at. The current layout groups the controls into three tabs, Overview (what is happen-ing), Filters (what to show and how far to project) and Areas (where to load data and what to alert on), and adds a one-time welcome card with three quick starts that puts a first-time visitor in front of either the Washington airspace or the Baltic shipping before they have to learn anything. On a phone the three panels and the map become four full-screen views behind a bottom tab bar, with an alert count on the Alerts tab, rather than a tall page with the map at the top and everything else stacked underneath.

7.4 Flight history and where it is going
Selecting an aircraft answers three questions at once. Its observed track is the positions this system has actually seen, seeded from the history already held for that target and then extended for as long as it stays selected, drawn as a solid bright line: it is what we watched, not what we were told. Its route legs are the published origin and destination, drawn as great circles from the origin airport to the aircraft's current position and on to the destination, with the airports marked and labelled. The panel adds the distance flown from the origin, the distance remaining and an arrival time at the current ground speed, and a button frames the whole flight.

Figure 2. A selected United flight from Philadelphia to Chicago O'Hare. The solid line behind the aircraft is its observed track; the dashed legs are the published route, interpolated as great circles from the origin airport through the aircraft to the destination. The rail lists distance flown, distance remain-ing and an arrival time at the current ground speed.

The legs are interpolated rather than drawn as straight lines, because a straight line between two airports is wrong on a Mercator projection: the shortest path curves. Sixty-five points along the great circle, using the standard intermediate-point formula [documented], make the drawn path the flown path. Longitudes are unwrapped as the path is built, so a transpacific route does not draw itself the long way around the world; there is a test for exactly that, on Tokyo to Los Angeles.

7.5 One layout bug worth recording
The endpoint includes a snippet of the upstream response body in its error detail, which is how the adsb.fi 403 was identified as a bot chal-lenge rather than a plain refusal. That text reached the interface verbatim, and a few hundred characters of nginx error page in a white-space: nowrap status chip widened the header enough to push the entire right-hand panel off screen. The fix has two halves: failures are summarized to a few words ("no source available: adsb.lol rate limited, adsb.fi blocked, opensky not responding") with the full text kept for the console, and the header, chips, layout grid and status bar are width-locked so no future content can widen the page whatever its origin. A regression test injects the exact string and asserts the panel is still on screen with zero document overflow. Any external text reaching page chrome is a layout hazard, and summarizing it is not sufficient alone.

8. Verification

Unit tests (26, no network). The geometry and the rules: haversine and destination round-tripping, ray casting against known points, inside and projected alerts, severity escalation with closing time, altitude band exclusion, a descending target entering the band mid-projection, the small-zone step-over case, a target tracking away raising nothing, emergency squawks, orbit detection needing both a full turn and a small footprint, target-kind filtering, worst-first ordering, the ground clamp, the advisory-zone rule, and the real FAA zone file loading with its pub-lished limits intact. Eight cover the close-approach math: a head-on pair whose TCPA must equal range over closing speed, a parallel pair that never closes, a crossing pair whose CPA is its offset, a pair already past, one alert per pair regardless of input order, a harbor of 30 moored

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ships raising nothing, the severity bands, and the pairwise pass appearing in the combined result. Two more check the great-circle path drawn for a route: that its summed length matches the direct distance and that it bows poleward of the chord, and that a path across the antimeridian stays continuous.

Two failed on first run and both were genuine: the unclamped altitude extrapolation of 5.2, and the great-circle convergence error in the test's own geometry in 5.1. A suite that passes entirely on first write is usually testing what the code does rather than what it should do.

Browser smoke test (Playwright). Loads the real page in headless Chromium, fails on any console or page error, waits for live targets, asserts nothing rendered lies outside the viewport, selects a vessel and asserts its detail renders visibly, injects the hostile error string and asserts the layout survives, draws a zone and asserts it produces alerts, switches region, and captures desktop and phone screenshots, against local de-velopment or production by URL. Two refinements made it trustworthy: MapLibre cancels in-flight tile requests whenever the view moves, which surfaces as dozens of ERR_ABORTED failures that are normal behavior, and a 502 from this system's own endpoint is the documented upstream refusal the page is supposed to explain, so the test asserts the explanation appears rather than treating it as fatal.

Production measurement. The Section 4 figures were taken against the deployed system, and so was the end state: 446 to 496 aircraft be-tween 2 and 12 seconds old via the relay, alongside 281 vessels [measured].

9. Operations and limitations

Deployment is one command with no build step, so what is in the repository is what runs, and the zone file is regenerated when the FAA chart cycle turns. The relay runs as a long-lived process wherever there is an ordinary IP, kept alive by forever, with a systemd user unit docu-mented for reboot persistence; its shared secret lives in a gitignored file the script reads itself, matching a Pages secret, and the relay end-points compare the bearer token byte by byte after a length check. For diagnosis the interface comes first: the status bar names the path that served the data, its age and the last poll, and the banner distinguishes a rate-limited edge from a stopped relay.

Known limits: AIS coverage is the Baltic and Gulf of Finland only, because that is what a keyless live feed covers; the projection is a straight line, right for a geofence warning and wrong for predicting behavior; NOTAM activation is not modeled, so a cold restricted area is still drawn and P-40's expansion during a presidential visit is not represented; AGL zones are compared against MSL altitude with terrain as the error; the relay is a dependency, and without it the aircraft feed degrades to intermittent; detection runs in the browser, so nothing is watching when no-body has the page open, which is why the engine is pure functions ready to move to a scheduled Worker; and adsb.fi's data is licensed for personal, non-commercial use [documented], which this project is, while commercial use would need their permission and adsb.lol's ODbL terms would need attention on redistribution.

10. Provenance ofthe load-bearing claims

CLAIM HOW IT IS KNOWN

ADS-B message formats and semantics RTCA DO-260B, ICAO Annex 10 Vol IV [standard]; JSON field semantics
from the readsb reference [documented]

alt_baro can be "ground"; 27 of 115 lacked track Direct inspection of live responses [measured]

AIS encodings, sentinels, ship types, packed ETA ITU-R M.1371 [standard]; ETA decoder validated live [measured]

Digitraffic needs gzip, 406 otherwise; Digitraffic-User raises
limits; CC BY 4.0

Digitraffic instructions [documented]; 406 reproduced across four
Accept values [measured]

adsb.lol is ODbL 1.0, open to everyone, limits load-dependent adsb.lol open data documentation [documented]

adsb.fi: 1 req/s, invalid responses may trigger IP blocks, non-commer-
cial, citation required, v2 deprecated for v3

adsb.fi opendata repository [documented]; v3 shape verified live [mea-sured]

Edge behavior: adsb.lol 429s, adsb.fi 403 challenge, OpenSky 522 Purpose-built diagnostic endpoint in production [measured]

Success rates 1 in 8, 3 in 8 with retries, 1 in 6 later Repeated polling of distinct viewports [measured]

No aggregator sends CORS headers; six alternative aggregators
unusable

Origin -bearing requests inspected, and each candidate probed di-rectly [measured]

D1 free tier: 5M read, 100k written per day; 50 queries per invocation Cloudflare D1 pricing and limits [documented]

FAA: 13 prohibited areas, 6,285-point circles, COUNTRY is "UNITED
STATES"

FAA SUA feature service queried directly [measured]

DC SFRA is 30 NM on the DCA VOR; prohibited areas under part 73 14 CFR 93 subpart V; 14 CFR 73 [standard]

Simplification 6,285 to 17 points; 506 vertices, 18.6 KB Output of the committed ingest script [measured]

Haversine and destination formulas Movable Type latitude/longitude reference [documented]

Palette contrast and color-vision figures Validator run against the page's dark surface, re-run after the rebrand
[measured]

Chrome palette and type are the other two sites' own tokens Read from the live jaronwilson.dev and jaronwilson.org stylesheets [mea-
sured]

55 alerts before advisory zones, 3 after Same view, before and after [measured]

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CLAIM HOW IT IS KNOWN

241 aircraft shown where the feed reported 1 Region switch with pruning disabled [measured]

End state: 446 to 496 aircraft, 2 to 12 s old, 281 vessels Public endpoints polled directly [measured]

CPA and TCPA are the standard measures, with alarms on a preset CPA
limit and TCPA warning time

IMO Resolution A.823(19), ARPA performance standards [standard]

Origin and destination are not in ADS-B and come from adsbdb; 404
means no scheduled route

adsbdb documentation [documented]; live callsigns resolved and 404s
observed [measured]

adsbdb answers the Cloudflare edge, unlike the ADS-B aggregators /api/route exercised from the deployed Worker [measured]

275 vessels gave 4 approach alerts at 1 NM, 27 at 5 NM; tightest 0.32
NM in 2m 14s

Detector run against live Baltic traffic [measured]

11. Author contributions

J.M.W. conceived the project, set its requirements and priorities, chose the platform (a static site with edge functions over a framework and an always-on server, when offered both), chose the relay over the alternatives when the egress problem was measured, decided the feature scope including the pinned tracking areas, per-feed pausing, vessel close-approach prediction and flight history, tested the deployed system throughout, reported the defects that led to several of the fixes described here, and reviewed the text.
Claude (Anthropic) implemented the software, ran the measurements reported as [measured], wrote the unit and browser tests, produced the figures, and drafted this paper, all under the direction of the first author. The provenance tags in the text and the table in Section 10 exist so that a reader can check any load-bearing claim against its source rather than trust either author.

12. Availability

The system is live at flysdown.pages.dev. The source, including the tests, the tools that regenerate the zone file, the figures and this docu-ment, is in the repository Jaron-Wilson/flysdown (private at the time of writing; contact the first author). Aircraft data is used under adsb.fi's personal, non-commercial terms with the required citation, and under adsb.lol's ODbL 1.0. Vessel data is Fintraffic Digitraffic, CC BY 4.0. Airspace geometry is the FAA's, in the public domain. Route data is from adsbdb (MIT). The basemap is OpenFreeMap, built on OpenStreetMap data, copyright OpenStreetMap contributors. Nothing here is for navigation.

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All URLs were retrieved and confirmed reachable on 17 September 2026. Two sources are cited from their own error responses rather than ren-dered documentation: adsb.fi's home page returns 403 to automated clients, which is the behavior described in Section 4, and the historical OpenSky REST documentation URL now returns 404 in favor of reference 8.

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